XUEWU CAI

Data Engineering & Analytics Platforms | Java, Python, SQL | AWS & Distributed Computing

Clarksville, MD | 410-446-6815 | gourmetcai@gmail.com | linkedin.com/in/xuewu-cai-a8069664 | U.S. citizen

PROFESSIONAL SUMMARY

Technology and financial engineering leader with more than 20 years across data-intensive financial platforms, quantitative research and trading systems. Built time-series infrastructure, market data feeds and Java database connectivity, and leads large-scale modeling analytics on AWS and IBM Symphony. Combines hands-on Java, Python, C/C++ and SQL database skills with technical leadership and enterprise model implementation.

CORE COMPETENCIES

Programming & Databases: Java, Python, C/C++, MATLAB, SQL, Oracle, SQL Server, Snowflake, MongoDB

Data & Integration: time-series infrastructure, market data feeds, database connectivity, financial model implementation

Cloud & Platforms: AWS, IBM Symphony, Docker, Spring Boot; large-scale computing and enterprise analytics

Delivery & Analysis: technical leadership, multi-project delivery, scenario analysis, forecasting; Copilot and Amazon Q

PROFESSIONAL EXPERIENCE

Director, Financial Engineering — Freddie Mac May 2014 – Present

McLean, VA | Credit Risk Transfer (STACR / ACIS)

- Direct development of modeling tools combining prepayment/default models, AWS and IBM Symphony computing, and Intex valuation analytics to assess bond performance and transaction economics.
- Own an enterprise forecasting platform for income statements, balance sheets and CECL estimates, supporting business planning, capital planning and DFAST stress testing.
- Lead engineering projects for STACR bond issuance and ACIS reinsurance within a CRT program protecting more than $1.6 trillion in mortgage loans.
- Deliver RPL and synthetic-structure deal memorandum, capital, economics and ROE analytics; use Copilot and harness engineering to support delivery.
- Manage the LIBOR-to-SOFR transition within CRT modeling tools and analytics.

Campbell & Company | Research Engineer and Engineering Roles Mar 2003 – Apr 2014

Washington, DC–Baltimore Area

Research Engineer, Portfolio and Risk Team (2009–2014); Quantitative Finance Engineer (2006–2008); Senior Software Engineer (2003–2006).

- Built time-series data infrastructure and market data feeds in MFC/C++, alongside a Java-based intraday cash equity trading system using the Caché database.
- Designed a Java connectivity layer unifying MATLAB access to production databases including Oracle, SQL Server, Caché and QADirect.
- Led a multithreaded, fault-tolerant Java/Portware/FIX trading platform; developed volatility/correlation forecasting models and risk reports using daily, intraday and tick data.

Senior Software Engineer — Aquilent Jun 1999 – Feb 2003

Java/Linux integration and data systems

- Built Java/Linux sensor integration systems with embedded Linux, JNI, serial communications and dynamic sensor discovery; developed Apache/Tomcat visualization portals and server-side satellite data components.

EDUCATION & PROFESSIONAL DEVELOPMENT

Master's Degree, Computer Science — University of Maryland Baltimore County

M.S., Computer Science & Engineering — Shanghai Jiao Tong University

Bachelor's Degree, Applied Mathematics — Dalian University of Technology

Professional development: CAIA (Chartered Alternative Investment Analyst) Level I

LANGUAGES

Chinese: native or bilingual proficiency | English: full professional proficiency